Math211
Discrete Mathematics

Proofs

Agenda

1.7 Introduction to Proofs

- direct proof, proof by contraposition, proof by contradiction

1.8 Proof by Cases ONLY

Discrete Mathematics and Its Applications

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Rules of Inference, Theorems and Proofs

- When is a mathematical argument correct?
- What techniques can we use to construct a mathematical argument?

- **Rules of Inference** – rules used in a proof to draw conclusions from assertions known to be true.
- **Theorem** – statement that can be shown to be true.
- **Proof** – sequence of statements, a valid argument, to show that a theorem is true.

Methods of Proof

1. Direct Proof
2. Indirect Proof
   A. Proof by Contraposition
   B. Proof by Contradiction
3. Proof by Cases

Even and Odd Integers

Even Integer

$$2 * (Any\ Integer) = even$$

if a is an even number, so you can write it as follows:

$$a = 2n, \qquad \text{where } n \text{ is integer}$$

Odd Integer

if a is an odd number, so you can write it as follows:

$$a = 2m + 1, \qquad \text{where } m \text{ is integer}$$

$$\text{Even} + 1 = \text{Odd}$$
$$\text{Odd} + 1 = \text{Even}$$

$$\neg\text{Even} = \text{Odd}$$
$$\neg\text{Odd} = \text{Even}$$

Perfect Square

Prefect Square

if a is a prefect square, so you can write it as follows:

$$\boxed{a = (n)^2, \qquad \text{where } n \text{ is integer}}$$

Rational and Irrational Numbers

Rational Number

if a is a *rational number*, so you can write it as follows:

$$a = \frac{n}{m}, \qquad \text{where } n, \text{and } m \text{ are integers}$$

$$\text{with NO common factor, and } m \neq 0$$

$$\neg\text{Rational} = \text{Irrational}$$

$$\neg\text{Irrational} = \text{Rational}$$

1.7.1. Direct Proof

1. Direct Proofs

- Most of the proofs we have seen so far are direct proofs.
- In a direct proof
  - You assume the hypothesis p is true
  - Give a direct series (sequence) of implications using the rules of inference as well as other results (proved independently) to show that the conclusion q holds.
- To perform a direct proof, assume that p is true, and show that q must therefore be true.

1. Direct Proofs

$p \to q$

1. We assume that p is true
2. We try to prove that q is also true
3. Then $p \to q$ is true.

Example 1 - direct proof

- Show that the square of an odd number is an odd number
- **Rephrased: if n is an odd integer, then n² is odd**

$$\text{"If } \underbrace{n \text{ is an odd integer}}_{p}, \text{ then } \underbrace{n^2 \text{ is odd}}_{q}."$$

$p \rightarrow q$

1. We assume that p is true
2. We try to prove that q is also true
3. Then $p \rightarrow q$ is true.

Example 1 - direct proof

$$\text{"If } \underbrace{n \text{ is an odd integer}}_{p}\text{, then } \underbrace{n^2 \text{ is odd}}_{q}\text{."}$$

1. We assume that p is true

$$n = 2m + 1, \qquad \text{where } m \text{ is integer.}$$

2. We try to prove that q is also true

$$\begin{aligned} n^2 &= (2m+1)^2 \\ &= 4m^2 + 4m + 1 \\ &= 2(2m^2 + 2m) + 1 \\ &= even \qquad\qquad + 1 = odd \end{aligned}$$

3. $\therefore p \to q$ is true.

Example 1 - direct proof

- **Show that the square of an odd number is an odd number**
- Rephrased: if n is an odd integer, then n^2 is odd

  p **q**

- Assume **the hypothesis (p) is true**, **n is odd**
  - Thus, $n = 2k+1$, for some k (definition of odd numbers)
  - $n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$
  - As n^2 is 2 times an integer + 1, **n^2 is thus odd**. **Therefore, the conclusion (q) is true.**
  - **Therefore, if n is an odd integer, then n^2 is odd**

Example 2 - direct proof

- **Show that the square of an even number is an even number**
- Rephrased: if n is even integer, then n^2 is even
  p **q**

- Assume **the hypothesis (p) is true**, **n is even**
  - Thus, $n = 2k$, for some k (definition of even numbers)
  - $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$
  - As n^2 is 2 times an integer, **n^2 is thus even**. **Therefore, the conclusion (q) is true.**
  - **Therefore, if n is an even integer, then n^2 is even**

Example 3 - direct proof

Prove that if n^3+5 is odd where n is an integer, then n is even

- Via direct proof
  - $n^3+5 = 2k+1$ for some integer k (definition of odd numbers)
  - $n^3 = 2k-4$
  - $n=\sqrt[3]{2k-4}$
  - Umm…

- So direct proof didn't work out. Next up: indirect proof

1.7.2. Proof by Contrapositive (Indirect Proof)

2. Proof by Contrapositive (Indirect Proofs)

- Consider an implication: $p \rightarrow q$
  - Recall that $(p \rightarrow q) \equiv (\neg q \rightarrow \neg p)$ (Its contrapositive)
  - You **assume that the conclusion is false ($\neg q$)**, then
  - Give a series of implications to show that such an assumption implies that the **premise is false ($\neg p$)**
  - Thus, show that if $\neg q$ is true, then $\neg p$ is true

To perform an indirect proof, do a direct proof on the contrapositive

2. Proof by Contrapositive (Indirect Proofs)

$$p \to q$$

$$\neg q \to \neg p$$

1. We assume that $\neg q$ is true
2. We try to prove that $\neg p$ is also true
3. Then $\neg q \to \neg p$ is true.
4. The $p \to q$ is also true.

Which to use?

- When do you use a direct proof versus an indirect proof?

- If it's not clear from the problem, try direct first, then indirect second
  - If indirect fails, try the other proofs

Example 1

Prove that if n^3+5 is odd where n is an integer, then n is even

- Via direct proof
  - $n^3+5 = 2k+1$ for some integer k (definition of odd numbers)
  - $n^3 = 2k-4$
  - n=$\sqrt[3]{2k-4}$
  - Umm…
- So direct proof didn't work out. Next up: indirect proof

Example 1

Prove that if n^3+5 is odd where n is an integer, then n is even

- Via indirect proof (Contrapositive)
  - **Contrapositive: If n is odd, then n^3+5 is even**
  - Assume n is odd, and show that n^3+5 is even
  - $n=2k+1$ for some integer k (definition of odd numbers)
  - $n^3+5 = (2k+1)^3+5 = 8k^3+12k^2+6k+6 = 2(4k^3+6k^2+3k+3)$
  - As $2(4k^3+6k^2+3k+3)$ is 2 times an integer, it is even.
  - **$\therefore \neg q \to \neg p$ is true, then $p \to q$ is also true.**

Example 2

Prove that If n^2 is an odd integer, then n is an odd integer

- Via indirect proof (Contrapositive)
  - **Contrapositive: If n is an even integer, then n^2 is an even integer**
  - Assume n is even, and show that n^2 is even
  - n=2k for some integer k (definition of even numbers)
  - $n^2 = (2k)^2 = 4k^2$
  - Since n^2 is 2 times an integer, it is even
  - **$\therefore \neg q \rightarrow \neg p$ is true, then $p \rightarrow q$ is also true.**

1.7.3. Proof by Contradiction (Indirect Proof)

3. Proof by Contradiction

1. Given a statement p, we want to prove p
    - Assume p is false and therefore $\neg p$ is true
    - Prove that $\neg p$ cannot occur (is false)
        - A contradiction exists

We want to prove p

- We Assume that:

(1) $p \to$ **F** therefore, $\neg p \to$ **T**

- We show that:

(1) $\neg p \to$ **F** (i.e., a **False** statement , say $r \wedge \neg r$)

(2) We conclude that $\neg p$ is false since (1) is **True** and therefore p is **True**.

3. Proof by Contradiction

2. Given a statement of the form p→q
   - To assume it's false, you only have to consider the case where p is true. and q is false

We want to show $p \rightarrow q$

(1) Assume the negation of the conclusion, i.e., $\neg q$

(2) Use it to show that $(p \wedge \neg q) \rightarrow \mathbf{F}$

(3) Since $((p \wedge \neg q) \rightarrow \mathbf{F}) \Leftrightarrow (p \rightarrow q)$ we are done

$$((p \wedge \neg q) \rightarrow F) \Leftrightarrow \neg(p \wedge \neg q)$$

$$\Leftrightarrow p \rightarrow q$$

Example 1 - Proof by Contradiction

Prove that if n is an integer and n^3+5 is odd, then n is even

- **Rephrased: If n^3+5 is odd, then n is even**
- **Thus, p is "n^3+5" is odd, q is "n is even"**

- Assume p and $\neg$q
  - Assume that n^3+5 is odd, and **n is odd**
- **Since n is odd:**
  - n=2k+1 for some integer k (definition of odd numbers)
  - $n^3+5 = (2k+1)^3+5 = 8k^3+12k^2+6k+6 = 2(4k^3+6k^2+3k+3)$
  - As $n^3+5 = 2(4k^3+6k^2+3k+3)$ is 2 times an integer, n^3+5 must be even. [We have reached contradiction n^3+5 is odd and n^3+5 is even].

Show that $(p \rightarrow q)$ by contradiction

Contradiction!

- We assumed p was true, and showed that this assumption implies that p must be false
- As p cannot be both true and false, we have reached our contradiction

Example 2- Proof by Contradiction

Prove that √2 is an irrational number

Let p the proposition "$\sqrt{2}$ is irrational".

1) To start a proof by contradiction, we suppose that $\neg p$ is true, then $\sqrt{2}$ is rational.
2) $\sqrt{2} = \frac{a}{b}$, where a and b are intergers without common factor and $b \neq 0$
3) $2 = \frac{a^2}{b^2}$
4) $a^2 = 2b^2$, then a is even, and you can write $a = 2m$, where m is integer.
5) $(2m)^2 = 2b^2$, then $4m^2 = 2b^2$.

Example 2- Proof by Contradiction

Prove that √2 is an irrational number

6) $b^2 = 2m^2$, then b is also even.

7) Because a and b are even, then a and b have a common factor 2.

8) Therefore, $\sqrt{2} = \frac{a}{b}$ is not rational, then $\neg p$ is false.

9) Therefore, p is true and $\sqrt{2}$ is irrational.

1.8.1 Proof by Cases

(WLOG) without loss of generalization

4. Proof by Cases

- Sometimes it is easier to prove a theorem by
  - Breaking it down into cases and
  - Proving each one separately
  - Assume the hypothesis is true and show that the conclusion is true for each case
- Show a statement is true by showing all possible cases are true
- To show that $(p_1 \vee p_2 \vee \ldots \vee p_n) \rightarrow q$
- We use the tautology

$$[(p_1 \vee p_2 \vee \ldots \vee p_n) \rightarrow q] \leftrightarrow [(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \ldots \wedge (p_n \rightarrow q)]$$

- A particular case of a proof by cases is an exhaustive proof in which all the cases are considered

Example 1 - Proof by cases

Prove "if n is an integer, then n^2+3n+2 is even."

- **Cases: n is odd, n is even**

Case 1: n is odd	**Case 2: n is even**
$n = 2k+1$ $= (2k+1)^2 + 3(2k+1) + 2$ $= 4k^2 + 4k + 1 + 6k + 3 + 2$ $= 4k^2 + 10k + 6$ $= 2(2k^2 + 5k + 3)$	$n = 2k$ $= (2k)^2 + 3(2k) + 2$ $= 4k^2 + 6k + 2$ $= 2(2k^2 + 3k + 1)$

Since both cases result in **even** values, then n^2+3n+2 is **even** if n is an integer.

Summarizing Proof techniques

Proof Technique	Approach to prove $p \to q$	Remarks
Direct Proof	Assume p, deduce q	The standard approach- usually the thing to try
Proof by Contraposition	Assume $\neg q$, derive $\neg p$	Use this $\neg q$ if as a hypothesis seems to give more ammunition then p would
Proof by Contradiction	Assume $p \land \neg q$, deduce a contradiction	Use this when q says something is not true
Proof by cases	Demonstrate $p \to q$ for all cases	May only be used for finite number of cases

Review Questions

Problem 1 - Proof by Contraposition

Prove that if a and b are integers, and a + b ≥ 15, then a ≥ 8 or b ≥ 8.

$$(a + b \geq 15) \rightarrow (a \geq 8) \vee (b \geq 8)$$

(Assume $\neg q$) Suppose $(a < 8) \wedge (b < 8)$.
(Show $\neg p$) Then $(a \leq 7) \wedge (b \leq 7)$,
and $(a + b) \leq 14$,
and $(a + b) < 15$.

Proof strategy:
Note that negation of conclusion is easier to start with here.

Problem 2 - Proof by Contraposition

Theorem

- **For n integer, if $3n + 2$ is odd, then n is odd.**
- **i.e. For n integer, $3n+2$ is odd $\rightarrow$ n is odd**

Again, negation of conclusion is easy to start with. Try direct proof ☹

Proof by Contraposition:

- Let p --- "$3n + 2$" is odd; q --- "n is odd"; we want to show that $p \rightarrow q$
- The contraposition of our theorem is $\neg q \rightarrow \neg p$
  - n is even $\rightarrow$ $3n + 2$ is even
- Now we can use a direct proof
- Assume $\neg q$, i.e, n is even therefore $n = 2k$ for some k
- Therefore $3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$ which is even.

Problem 3 - Proof by Contraposition

Prove that if the square of an integer is odd, then the integer must be odd

- Hence, prove n^2 odd $\rightarrow$ n odd
- To do a proof by contraposition prove n even $\rightarrow$ n^2 even
- If n is even, then it can be written as $2m$ where m is an integer, i.e. even/odd.
- Then, $n^2 = 4m^2$ which is always even no matter what m^2 is because of the factor 4.

Problem 4 - Proof by Contradiction

- **Theorem "If 3n+2 is odd, then n is odd"**
- **Let p = "3n+2 is odd" and q = "n is odd"**

1. Assume p and ¬q i.e., 3n+2 is odd and n is not odd
2. Because n is not odd, it is even
3. if n is even, n = 2k for some k, and therefore 3n+2 = 3 (2k) + 2 = 2 (3k + 1), which is even
4. So, we have a contradiction, 3n+2 is odd and 3n+2 is even.

- Therefore, we conclude $p \to q$, i.e., "If 3n+2 is odd, then n is odd"

Next Lecture

- Set theory
  - set builder notation, subset, Cartesian product, power set, union, intersection, complements, set identities.

See you next lecture